

H524 Introduction to Biostatistics

Week 5: Hypothesis Testing

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Quick Review: Confidence Intervals

Last week we covered:

- Point estimates vs interval estimates
- Confidence intervals for population mean
- General form: $\bar{x} \pm t_{\alpha/2, n-1} \times \frac{s}{\sqrt{n}}$
- Interpretation: 95% of intervals will contain μ
- Trade-off between confidence level and precision

This Week: Hypothesis testing - making decisions about population parameters

Reading: Pagano & Gauvreau Chapter 10

From Estimation to Testing

Two approaches to statistical inference:

1. Estimation (Weeks 3-4):

- What is the value of the parameter?
- Construct confidence interval
- Example: “True mean cholesterol is between 195 and 205 mg/dL”

2. Hypothesis Testing (This week):

- Is a specific claim about the parameter true?
- Make a decision: reject or fail to reject
- Example: “Is the true mean cholesterol greater than 200 mg/dL?”

Both use: Sample data, sampling distributions, probability theory

What is a Hypothesis Test?

Definition: A statistical procedure to assess evidence about a population parameter

Basic idea:

- 1 Make a claim about a population parameter
- 2 Collect sample data
- 3 Use probability to decide if data supports the claim
- 4 Make a decision: reject or fail to reject the claim

Medical example:

- Claim: “New drug lowers blood pressure by more than 10 mmHg on average”
- Collect data from clinical trial
- Calculate probability of observing this data if drug has no effect
- If probability is very small, reject “no effect” claim

The Logic of Hypothesis Testing

Key principle: *Proof by contradiction*

Analogy - Criminal trial:

- **Null hypothesis:** Defendant is innocent (default assumption)
- **Alternative hypothesis:** Defendant is guilty (what we're trying to prove)
- **Evidence:** Trial data
- **Decision:** Guilty (reject null) or Not Guilty (fail to reject null)
- **Standard:** Beyond reasonable doubt (low probability threshold)

Important: “Not guilty” $\neq$ “innocent”

Similarly: “Fail to reject H_0 ” $\neq$ “ H_0 is true”

We never “accept” the null hypothesis - we only fail to reject it!

Null and Alternative Hypotheses

Null Hypothesis (H_0):

- Statement of “no effect” or “no difference”
- Status quo, default assumption
- What we assume is true unless proven otherwise
- Always contains equality: $=$, $\leq$, or $\geq$

Alternative Hypothesis (H_A or H_1):

- Statement of “effect exists” or “difference exists”
- What the researcher wants to demonstrate
- What we conclude if evidence is strong enough
- Contains inequality: $\neq$, $>$, or $<$

Key rule: H_0 and H_A must be:

- Mutually exclusive (can't both be true)
- Exhaustive (one must be true)

Types of Hypotheses: Two-Sided vs One-Sided

Two-sided (two-tailed) test:

$$H_0 : \mu = \mu_0 \quad \text{vs} \quad H_A : \mu \neq \mu_0$$

- Testing if parameter differs from null value (either direction)
- Example: “Is mean cholesterol different from 200 mg/dL?”
- Most common, more conservative

One-sided (one-tailed) test - Upper tail:

$$H_0 : \mu \leq \mu_0 \quad \text{vs} \quad H_A : \mu > \mu_0$$

- Testing if parameter is greater than null value
- Example: “Is mean blood pressure higher than 120 mmHg?”

One-sided test - Lower tail:

$$H_0 : \mu \geq \mu_0 \quad \text{vs} \quad H_A : \mu < \mu_0$$

- Testing if parameter is less than null value
- Example: “Does drug reduce cholesterol below 200 mg/dL?”

Example: Setting Up Hypotheses

Scenario 1: Normal body temperature is claimed to be 98.6°F. We want to test if this is accurate.

$$H_0 : \mu = 98.6 \quad \text{vs} \quad H_A : \mu \neq 98.6$$

Scenario 2: A new drug claims to lower blood pressure. Current mean is 140 mmHg.

$$H_0 : \mu \geq 140 \quad \text{vs} \quad H_A : \mu < 140$$

Scenario 3: We suspect exposure to a toxin increases cancer rate above baseline 5%.

$$H_0 : p \leq 0.05 \quad \text{vs} \quad H_A : p > 0.05$$

Key decision: Choose one-sided vs two-sided *before* collecting data, based on research question!

The Test Statistic

Definition: A value calculated from sample data that measures how far the sample is from the null hypothesis

General form:

$$\text{Test Statistic} = \frac{\text{Estimate} - \text{Null Value}}{\text{Standard Error}}$$

For testing mean (large sample or σ known):

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

For testing mean (σ unknown, small sample):

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

where μ_0 is the null hypothesis value

Interpreting the Test Statistic

What does the test statistic tell us?

- Measures “how many standard errors away” the sample mean is from null value
- Large $|t|$ or $|Z|$: Sample mean far from null value
- Small $|t|$ or $|Z|$: Sample mean close to null value

Examples:

- $t = 0.5$: Sample mean is 0.5 SE from null (not unusual)
- $t = 2.5$: Sample mean is 2.5 SE from null (unusual!)
- $t = -3.8$: Sample mean is 3.8 SE below null (very unusual!)

Question: How large does $|t|$ need to be for us to reject H_0 ?

Answer: Depends on:

- Degrees of freedom (sample size)
- Significance level (how much risk we'll accept)
- One-sided vs two-sided test

The P-value

Definition: The probability of observing a test statistic as extreme as (or more extreme than) what we observed, *assuming H_0 is true*

For two-sided test:

$$p\text{-value} = \Pr(|T| \geq |t_{obs}| \mid H_0 \text{ true})$$

For one-sided test (upper tail):

$$p\text{-value} = \Pr(T \geq t_{obs} \mid H_0 \text{ true})$$

For one-sided test (lower tail):

$$p\text{-value} = \Pr(T \leq t_{obs} \mid H_0 \text{ true})$$

Interpretation:

- Small p-value: Observed data would be very unlikely if H_0 were true
- Large p-value: Observed data consistent with H_0

Understanding P-values

Common misconceptions:

- $\times$ p-value is NOT $\Pr(H_0 \text{ is true})$
- $\times$ p-value is NOT $\Pr(\text{data due to chance})$
- $\times$ $1 - \text{p-value}$ is NOT $\Pr(H_A \text{ is true})$

Correct interpretation:

“If the null hypothesis were true, the probability of obtaining a test statistic as extreme as or more extreme than the one we observed is [p-value]”

Rule of thumb for interpretation:

- $p < 0.01$: Very strong evidence against H_0
- $0.01 \leq p < 0.05$: Strong evidence against H_0
- $0.05 \leq p < 0.10$: Weak evidence against H_0
- $p \geq 0.10$: Little or no evidence against H_0

Significance Level (α)

Definition: The probability threshold for rejecting H_0

Decision rule:

- If $p\text{-value} < \alpha$: Reject H_0 (statistically significant)
- If $p\text{-value} \geq \alpha$: Fail to reject H_0 (not statistically significant)

Common choices:

- $\alpha = 0.05$ (most common in biostatistics)
- $\alpha = 0.01$ (more stringent, reduces false positives)
- $\alpha = 0.10$ (more lenient, exploratory studies)

Important: Choose α *before* collecting data!

Changing α after seeing results is called “p-hacking” and invalidates the test

Two Types of Errors

Hypothesis testing is a decision with uncertainty

		Reality (Unknown)	
		H_0 True	H_0 False
Decision	Reject H_0	Type I Error (α)	Correct (Power)
	Fail to Reject	Correct ($1 - \alpha$)	Type II Error (β)

Type I Error: Reject H_0 when it's actually true (false positive)

- Probability = α (significance level)
- Example: Conclude drug works when it doesn't

Type II Error: Fail to reject H_0 when it's actually false (false negative)

- Probability = β
- Example: Miss a real drug effect

Type I and Type II Errors: Medical Examples

Example 1: Disease screening test

H_0 : Patient does not have disease

H_A : Patient has disease

- **Type I Error:** Diagnose disease when patient is healthy (false positive)
 - Consequence: Unnecessary treatment, anxiety
- **Type II Error:** Miss disease when patient is sick (false negative)
 - Consequence: Delayed treatment, disease progression

Example 2: Drug approval

H_0 : Drug has no effect

H_A : Drug is effective

- **Type I Error:** Approve ineffective drug
 - Consequence: Patients receive useless treatment
- **Type II Error:** Reject effective drug
 - Consequence: Beneficial treatment unavailable

Controlling Type I Error

Type I Error rate = α (significance level)

We control Type I error by choosing α :

- Set $\alpha = 0.05$: 5% chance of false positive
- Set $\alpha = 0.01$: 1% chance of false positive (more conservative)
- Set $\alpha = 0.10$: 10% chance of false positive (more lenient)

Trade-off:

- Decreasing α (stricter) reduces Type I errors
- But increases Type II errors (β) - harder to detect real effects
- Can't make both arbitrarily small without increasing sample size

Convention in biostatistics:

- Typically prioritize controlling Type I error
- $\alpha = 0.05$ is standard (sometimes 0.01 for critical decisions)
- Better to miss a real effect than falsely claim one exists

Statistical Power

Power = Probability of correctly rejecting H_0 when H_A is true

$$\text{Power} = 1 - \beta$$

What affects power?

- 1 **Sample size (n):** Larger $n \Rightarrow$ higher power
- 2 **Effect size:** Larger true difference $\Rightarrow$ higher power
- 3 **Significance level (α):** Larger $\alpha \Rightarrow$ higher power
- 4 **Variability (σ):** Lower $\sigma \Rightarrow$ higher power

Typical goal: Power ≥ 0.80 (80%)

Means 80% chance of detecting the effect if it truly exists

Power analysis: Calculate required sample size to achieve desired power

- Done *before* collecting data
- Ensures study is adequately sized to detect meaningful effects

One-Sample t-Test

Used when:

- Testing hypothesis about a population mean μ
- Population SD (σ) is unknown
- Sample is random and approximately normal (or n large by CLT)

Hypotheses:

Two-sided: $H_0 : \mu = \mu_0$ vs $H_A : \mu \neq \mu_0$

One-sided: $H_0 : \mu \leq \mu_0$ vs $H_A : \mu > \mu_0$ (or opposite)

Test statistic:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Under H_0 : $t \sim t_{n-1}$ (t-distribution with $n - 1$ degrees of freedom)

Decision:

- Calculate p-value from t-distribution
- Reject H_0 if $p < \alpha$

Example: Body Temperature

Question: Is mean body temperature really 98.6°F?

Data: Sample of $n = 130$ healthy adults

- Sample mean: $\bar{x} = 98.25^\circ\text{F}$
- Sample SD: $s = 0.73^\circ\text{F}$

Step 1: Set up hypotheses

$$H_0 : \mu = 98.6 \quad \text{vs} \quad H_A : \mu \neq 98.6$$

Step 2: Calculate test statistic

$$t = \frac{98.25 - 98.6}{0.73/\sqrt{130}} = \frac{-0.35}{0.064} = -5.47$$

Step 3: Find p-value

From t-distribution with $df = 129$: $p < 0.0001$ (two-sided)

Step 4: Conclusion

$p < 0.05$, so reject H_0 . Strong evidence that mean body temperature is not 98.6°F.

Example: Blood Pressure Drug

Question: Does new drug lower systolic BP below 140 mmHg?

Data: $n = 25$ patients after treatment

- Sample mean: $\bar{x} = 135$ mmHg
- Sample SD: $s = 12$ mmHg

Step 1: Set up hypotheses (one-sided)

$$H_0 : \mu \geq 140 \quad \text{vs} \quad H_A : \mu < 140$$

Step 2: Calculate test statistic

$$t = \frac{135 - 140}{12/\sqrt{25}} = \frac{-5}{2.4} = -2.08$$

Step 3: Find p-value (one-sided, lower tail)

From t_{24} distribution: $p = 0.024$

Step 4: Conclusion

$p = 0.024 < 0.05$, reject H_0 . Evidence that drug lowers BP below 140 mmHg.

Critical Values vs P-values

Two equivalent approaches to hypothesis testing:

Method 1: P-value approach (modern, preferred)

- 1 Calculate test statistic t
- 2 Calculate p-value
- 3 Compare p-value to α
- 4 Reject H_0 if $p < \alpha$

Method 2: Critical value approach (classical)

- 1 Calculate test statistic t
- 2 Find critical value(s) from t-table for given α
- 3 Compare $|t|$ to critical value
- 4 Reject H_0 if $|t| > t_{critical}$ (two-sided)

Advantage of p-values:

- Provides exact strength of evidence
- Can compare to any α level
- Software gives p-values automatically

Critical Values for Two-Sided Tests

For two-sided test at $\alpha = 0.05$:

Reject H_0 if $|t| > t_{0.975, n-1}$

df	$t_{0.975}$	Interpretation
5	2.571	$ t > 2.571$ is significant
10	2.228	$ t > 2.228$ is significant
20	2.086	$ t > 2.086$ is significant
30	2.042	$ t > 2.042$ is significant
100	1.984	$ t > 1.984$ is significant
∞	1.960	$ t > 1.960$ is significant

Note: As df increases, t critical values approach $Z = 1.96$

For large samples ($n > 30$), can use normal approximation

One-Sample z-Test

Used when:

- Testing hypothesis about population mean μ
- Population SD (σ) is **known** (rare in practice!)
- OR sample size is very large ($n > 100$)

Test statistic:

$$Z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

Under H_0 : $Z \sim N(0, 1)$ (standard normal)

Critical values for $\alpha = 0.05$:

- Two-sided: $|Z| > 1.96$
- One-sided: $Z > 1.645$ (upper) or $Z < -1.645$ (lower)

In practice: Almost always use t-test instead

- Rarely know σ without knowing μ
- For large n , t-test and z-test give nearly identical results

Example: IQ Test with Known SD

Scenario: IQ tests are designed to have $\sigma = 15$ (known). We want to test if a special program increases mean IQ above 100.

Data: $n = 64$ students in program

- Sample mean: $\bar{x} = 103$
- Known SD: $\sigma = 15$

Hypotheses:

$$H_0 : \mu \leq 100 \quad \text{vs} \quad H_A : \mu > 100$$

Test statistic:

$$Z = \frac{103 - 100}{15/\sqrt{64}} = \frac{3}{1.875} = 1.60$$

P-value: $\Pr(Z > 1.60) = 0.055$

Conclusion: $p = 0.055 > 0.05$, fail to reject H_0 . Marginal evidence that program increases IQ.

Relationship Between CIs and Hypothesis Tests

Key insight: Confidence intervals and hypothesis tests are closely related!

For two-sided test at level α :

Reject $H_0 : \mu = \mu_0$ if and only if the $(1 - \alpha) \times 100\%$ CI does NOT contain μ_0

Example: Blood pressure drug data

- $n = 25$, $\bar{x} = 135$ mmHg, $s = 12$ mmHg
- 95% CI: $135 \pm 2.064 \times \frac{12}{\sqrt{25}} = (130.0, 140.0)$
- Test $H_0 : \mu = 140$ vs $H_A : \mu \neq 140$
- Since 140 is *in* the CI, we would *fail to reject* H_0 (for two-sided test)

Note: This equivalence only holds for two-sided tests!

For one-sided tests, use one-sided confidence bounds

Using CIs for Multiple Comparisons

Advantage of confidence intervals:

Can simultaneously test multiple null values!

Example: Cholesterol drug trial

- Sample: $n = 50$, $\bar{x} = 185$ mg/dL, $s = 30$ mg/dL
- 95% CI: (176.5, 193.5) mg/dL

We can immediately conclude:

- $\mu = 200$? Not in CI, reject H_0 ($p < 0.05$)
- $\mu = 190$? In CI, fail to reject H_0 ($p > 0.05$)
- $\mu = 180$? In CI, fail to reject H_0 ($p > 0.05$)
- $\mu = 170$? Not in CI, reject H_0 ($p < 0.05$)

Interpretation: Any value inside the CI is “plausible” given the data
Any value outside is “implausible” at the $\alpha = 0.05$ level

Statistical vs Clinical Significance

Statistical significance: $p < \alpha$

Clinical significance: Meaningful in practice

These are NOT the same!

Example 1: Large sample, small effect

- Drug lowers BP by 2 mmHg on average
- With $n = 10000$: $p < 0.001$ (highly significant)
- But 2 mmHg reduction may not be clinically meaningful

Example 2: Small sample, large effect

- Drug lowers BP by 15 mmHg on average
- With $n = 10$: $p = 0.08$ (not significant at $\alpha = 0.05$)
- But 15 mmHg reduction is clinically important!
- Need larger sample to demonstrate statistical significance

Always consider: Effect size, confidence interval width, clinical context

Sample Size and Hypothesis Testing

Effect of sample size on statistical tests:

Larger samples ($n \uparrow$):

- Smaller standard error: $SE = s/\sqrt{n}$
- Larger test statistics: $t = \frac{\bar{x} - \mu_0}{SE}$
- Smaller p-values (easier to reject H_0)
- Higher power to detect small effects
- Narrower confidence intervals

Implication:

- With large enough n , even tiny effects become “significant”
- Always report:
 - Effect size (e.g., difference in means)
 - Confidence interval
 - Clinical/practical importance
- Don't just report “ $p < 0.05$ ”!

Common Mistakes in Hypothesis Testing

1. Confusing “fail to reject” with “accept”

- ✗ “We accept the null hypothesis”
- ✓ “We fail to reject the null hypothesis”
- Absence of evidence $\neq$ evidence of absence

2. Choosing one-sided vs two-sided after seeing data

- Must decide before collecting data based on research question

3. Changing α after seeing p-value

- “ $p = 0.06$, let's use $\alpha = 0.10$ ” is p-hacking!

4. Interpreting p-value as $\Pr(H_0 \text{ true})$

- P-value is $\Pr(\text{data} \mid H_0)$, not $\Pr(H_0 \mid \text{data})$

5. Ignoring assumptions

- t-test assumes approximate normality (or large n)
- Check with histogram, Q-Q plot, or normality tests

6. Confusing statistical and practical significance

- Always consider effect size and clinical importance

Checking Assumptions

One-sample t-test assumes:

- 1 Random sample from population
- 2 Observations are independent
- 3 Population is approximately normal (or n large)

How to check normality:

1. Visual methods:

- Histogram: Should be roughly bell-shaped
- Q-Q plot: Points should follow straight line
- Boxplot: Should be roughly symmetric

2. Formal tests:

- Shapiro-Wilk test: Tests H_0 : data is normal
- Anderson-Darling test
- But: With large n , minor deviations become “significant”

3. Central Limit Theorem:

- For $n \geq 30$, t-test is robust to non-normality
- Exception: Extreme outliers or strong skewness

One-Sample t-Test in R

Example: Body temperature data

```
# Body temperature data
temps <- c(98.0, 98.2, 98.4, 98.6, 98.0, 97.8, 98.8,
          98.0, 98.4, 98.6, 98.5, 98.3)

# Descriptive statistics
mean(temps)      # 98.30
sd(temps)        # 0.311
length(temps)    # 12

# Two-sided test: H0:  $\mu = 98.6$  vs HA:  $\mu \neq 98.6$ 
t.test(temps, mu = 98.6, alternative = "two.sided")

# Output includes:
#   t-statistic, degrees of freedom, p-value
#   95% confidence interval, sample mean
```


One-Sided t-Test in R (Part 1 - Setup)

```
# Blood pressure data (after treatment)
bp <- c(138, 142, 135, 140, 132, 128, 145, 136,
        139, 133, 141, 137, 134, 130, 143)

# One-sided test: H0:  $\mu \geq 140$  vs HA:  $\mu < 140$ 
result <- t.test(bp, mu = 140, alternative = "less")

# Extract components
result$statistic      # t-statistic
result$parameter      # degrees of freedom
result$p.value        # p-value
result$conf.int       # one-sided confidence bound
```

One-Sided t-Test in R (Part 2 - Manual Calculation)

```
# Manual calculation
xbar <- mean(bp)
s <- sd(bp)
n <- length(bp)
t_stat <- (xbar - 140) / (s / sqrt(n))
p_value <- pt(t_stat, df = n-1)

cat("Sample mean:", round(xbar, 2), "\n")
cat("t-statistic:", round(t_stat, 3), "\n")
cat("p-value:", round(p_value, 4), "\n")
```

Checking Normality in R (Part 1 - Visual Checks)

```
# Generate some data
set.seed(524)
data <- rnorm(30, mean = 100, sd = 15)

# Visual checks
par(mfrow = c(1, 3))

# 1. Histogram
hist(data, breaks = 10, main = "Histogram",
      xlab = "Value", col = "lightblue")

# 2. Q-Q plot
qqnorm(data, main = "Q-Q Plot")
qqline(data, col = "red", lwd = 2)
```

Checking Normality in R (Part 2 - Tests)

```
# 3. Boxplot
boxplot(data, main = "Boxplot", ylab = "Value",
        col = "lightgreen")

par(mfrow = c(1, 1))

# Shapiro-Wilk test
shapiro.test(data)

# p > 0.05: fail to reject normality (data looks normal)
```

Power and Sample Size in R

```
# Calculate power for one-sample t-test
library(pwr) # install.packages("pwr")

# Power for n = 30 (effect size d = 0.5)
pwr.t.test(n = 30, d = 5/10, sig.level = 0.05,
           type = "one.sample",
           alternative = "two.sided")

# Power approx 0.70

# Sample size needed for 80% power
pwr.t.test(d = 5/10, power = 0.80,
           sig.level = 0.05,
           type = "one.sample",
           alternative = "two.sided")

# n approx 34
```

Complete Hypothesis Test Example (Part 1 - Setup)

```
# Research question: Is mean cholesterol > 200 mg/dL?
chol <- c(215, 198, 203, 225, 192, 210, 208, 195,
          218, 202, 212, 205, 220, 198, 207, 213,
          201, 209, 217, 204)

# Step 1: Descriptive statistics
cat("Sample size:", length(chol), "\n")
cat("Sample mean:", round(mean(chol), 2), "\n")
cat("Sample SD:", round(sd(chol), 2), "\n\n")

# Step 2: Check assumptions
shapiro.test(chol) # Test normality

# Step 3: Conduct test (one-sided, upper tail)
result <- t.test(chol, mu = 200,
                  alternative = "greater",
                  conf.level = 0.95)

print(result)
```

Complete Hypothesis Test Example (Part 2 - Interpret)

```
# Step 4: Interpret
if(result$p.value < 0.05) {
  cat("\nReject H0. Evidence that mean cholesterol > 200.\n")
} else {
  cat("\nFail to reject H0. Insufficient evidence.\n")
}

# Step 5: Effect size
effect_size <- (mean(chol) - 200) / sd(chol)
cat("Cohen's d =", round(effect_size, 3), "\n")
```

AI for Hypothesis Testing

How AI can help this week:

- **Conceptual clarification:** Understanding p-values, Type I/II errors
- **Test selection:** Choosing appropriate test for your data
- **R code generation:** Conducting t-tests, checking assumptions
- **Interpretation:** Understanding output from `t.test()`
- **Power analysis:** Sample size calculations
- **Visualization:** Creating plots to check assumptions

Example prompts:

- “Explain the difference between Type I and Type II errors with a medical example”
- “Show R code to conduct one-sided t-test testing if mean ≥ 100 ”
- “How do I interpret a p-value of 0.03?”
- “Create Q-Q plot in R to check normality assumption”

AI Demo: Understanding P-values (Part 1)

Prompt: "Explain what a p-value of 0.02 means in plain language"

Claude AI Response:

A p-value of 0.02 means:

"If the null hypothesis were actually true, we would expect to see a result as extreme as (or more extreme than) what we observed only 2% of the time by random chance alone."

In practical terms:

- This is considered statistically significant ($p < 0.05$)
- It provides evidence against the null hypothesis
- There's only a 2% chance of getting this result if the null hypothesis is true
- We would typically reject the null hypothesis

AI Demo: Understanding P-values (Part 2)

Important: This does NOT mean:

- There's a 2% chance the null hypothesis is true
- There's a 98% chance our result is "real"
- The effect is necessarily large or important

Always consider the effect size and clinical significance in addition to the p-value.

AI Demo: Test Selection

Prompt: “I have data from 20 patients before and after treatment. What test should I use?”

Claude AI Response:

For before-and-after data on the SAME patients, you should use a PAIRED t-test (not a one-sample t-test).

Here's why:

- You have two measurements per patient (before and after)
- The measurements are dependent (same person)
- You're interested in the change within individuals

R code:

```
before <- c(...) # before measurements  
after <- c(...)  # after measurements
```

```
# Method 1: Paired t-test directly  
t.test(after, before, paired = TRUE)
```

```
# Method 2: Create difference variable  
difference <- after - before  
t.test(difference, mu = 0) # Test if mean difference != 0
```

Both methods give identical results. The paired t-test is more powerful than comparing groups independently because it accounts for individual variation.

AI Demo: Checking Assumptions (Part 1 - Visual)

Prompt: “How do I check if my data is normal enough for a t-test?”

Claude AI Response:

```
# Check normality for t-test
data <- c(...) # your data

# Visual checks (most important!)
par(mfrow = c(2, 2))

# 1. Histogram
hist(data, breaks = 10, main = "Histogram", prob = TRUE)
curve(dnorm(x, mean = mean(data), sd = sd(data)),
      add = TRUE, col = "red", lwd = 2)

# 2. Q-Q plot (most useful!)
qqnorm(data)
qqline(data, col = "red")
# Points should roughly follow the line

# 3. Boxplot
boxplot(data, main = "Boxplot")
# Should be roughly symmetric, check for outliers
```

AI Demo: Checking Assumptions (Part 2 - Tests)

```
# 4. Density plot
plot(density(data), main = "Density Plot")

# Formal test
shapiro.test(data)
# p > 0.05 suggests normality
# But with large n, minor deviations become "significant"

# Rule of thumb: t-test is robust for n >= 30 unless
# extreme outliers or strong skewness
```

AI Best Practices for Hypothesis Testing

DO:

- Ask AI to explain concepts
- Request code examples
- Verify calculations independently
- Check assumptions visually
- Ask for interpretation help
- Learn from AI explanations

DON'T:

- Blindly trust AI p-values
- Skip checking assumptions
- Use AI without understanding
- Choose test based on desired result
- Ignore effect sizes
- Forget clinical context

Critical thinking points:

- 1 Does the test AI suggested match your research question?
- 2 Are assumptions reasonable for your data?
- 3 Does the interpretation make practical sense?
- 4 Is statistical significance the same as clinical importance?

Summary: Hypothesis Testing Framework

Complete hypothesis testing procedure:

- 1 **State hypotheses:** H_0 vs H_A (before collecting data!)
- 2 **Choose significance level:** Usually $\alpha = 0.05$
- 3 **Check assumptions:** Normality, independence, random sampling
- 4 **Calculate test statistic:** $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$
- 5 **Find p-value:** From t-distribution with $n - 1$ df
- 6 **Make decision:** Reject H_0 if $p < \alpha$
- 7 **State conclusion:** In context of research question
- 8 **Report effect size:** Difference, confidence interval, practical importance

Remember:

- Statistical significance $\neq$ practical importance
- Fail to reject $\neq$ accept
- Always consider Type I and Type II error consequences
- Report confidence intervals alongside p-values

Key Concepts to Remember

Hypotheses:

- H_0 : null hypothesis (status quo, no effect)
- H_A : alternative hypothesis (what we want to show)
- Two-sided vs one-sided tests

Decision-making:

- P-value: probability of data or more extreme, given H_0 true
- α : significance level (Type I error rate)
- Reject H_0 if $p < \alpha$

Errors:

- Type I (α): False positive (reject true H_0)
- Type II (β): False negative (fail to reject false H_0)
- Power = $1 - \beta$: Probability of detecting true effect

Tests covered:

- One-sample t-test: $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$
- One-sample z-test: $Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ (rare)

Week 6: Two-Sample Tests

We'll extend hypothesis testing to compare two groups:

- Two-sample t-test (independent groups)
- Paired t-test (dependent samples)
- Confidence intervals for difference in means
- Assumptions and diagnostics
- Effect sizes (Cohen's d)

Examples:

- Treatment vs control groups
- Before vs after measurements
- Drug A vs Drug B comparison

Reading: Chapter 11

Thank you!

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